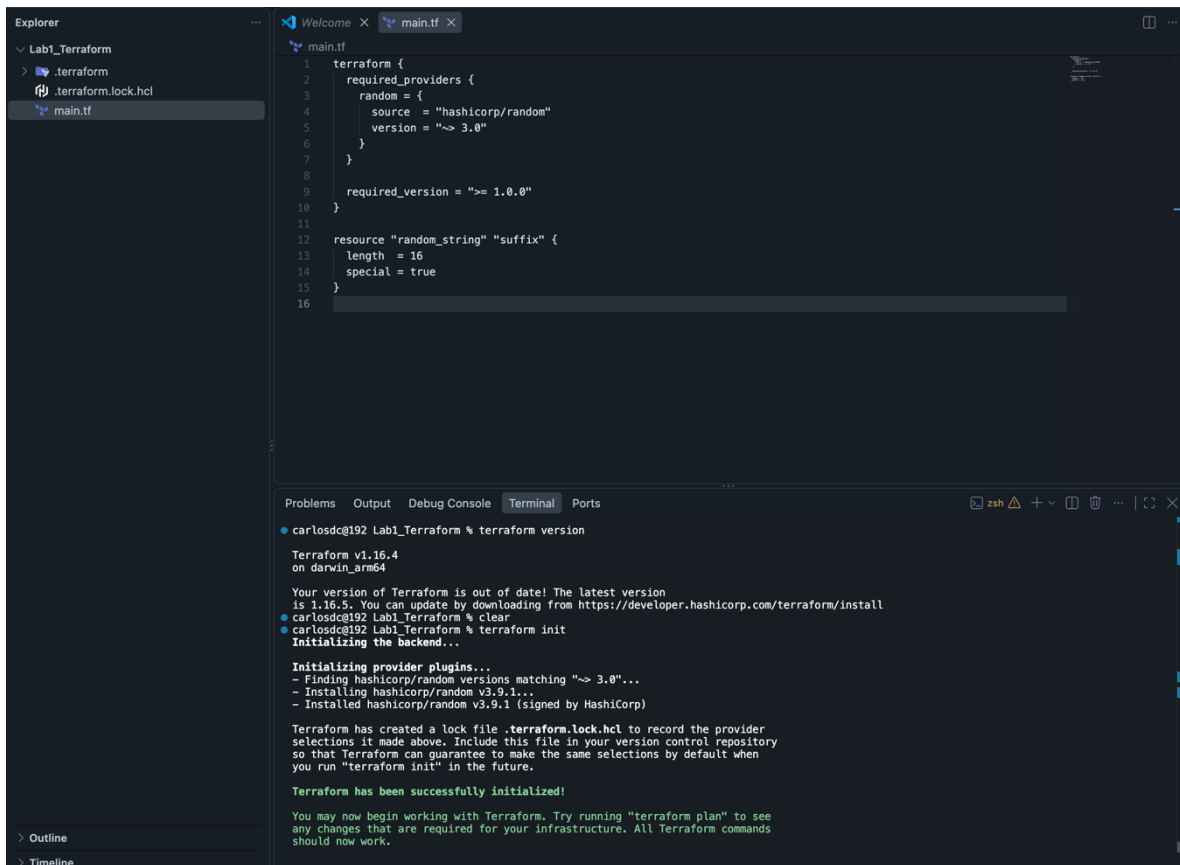


Terraform 1



The screenshot shows a Visual Studio Code editor with a Terraform configuration file named `main.tf` in the Explorer pane. The configuration defines a required provider and a resource. The terminal pane shows the execution of `terraform version`, `terraform init`, and `terraform validate`.

```
1 terraform {
2   required_providers {
3     random = {
4       source = "hashicorp/random"
5       version = "~> 3.0"
6     }
7   }
8   required_version = ">= 1.0.0"
9 }
10
11
12 resource "random_string" "suffix" {
13   length = 16
14   special = true
15 }
16
```

Terminal Output:

```
carlosdc@192 Lab1_Terraform % terraform version
Terraform v1.16.4
on darwin_arm64

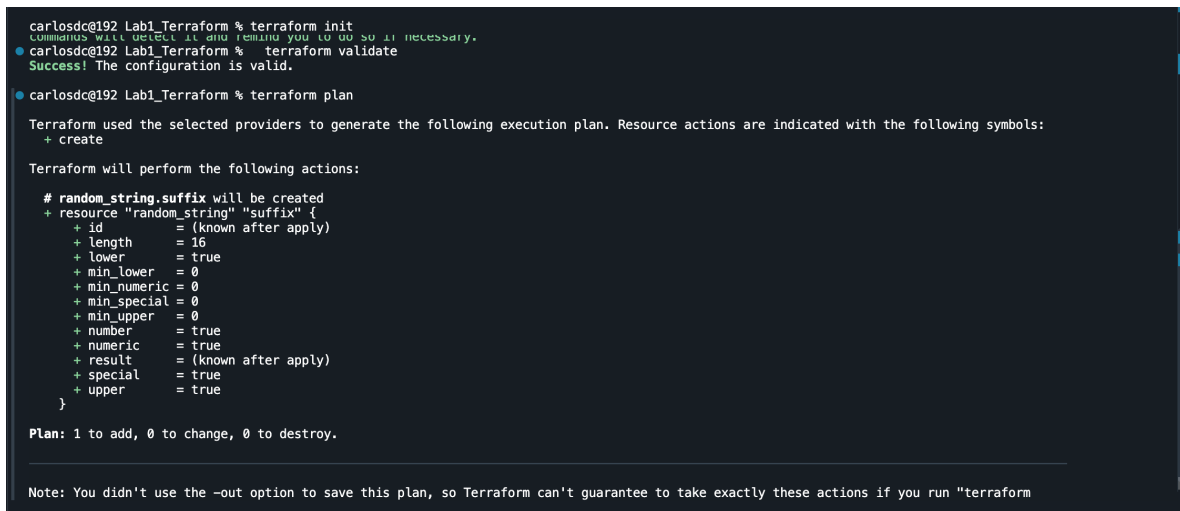
Your version of Terraform is out of date! The latest version
is 1.16.5. You can update by downloading from https://developer.hashicorp.com/terraform/install
carlosdc@192 Lab1_Terraform % terraform init
Initializing the Backend...

Initializing provider plugins...
- Finding hashicorp/random versions matching "~> 3.0"...
- Installing hashicorp/random v3.9.1...
- Installed hashicorp/random v3.9.1 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.
```



The screenshot shows the terminal output of the `terraform plan` command. It displays the execution plan for the `random_string.suffix` resource, indicating that it will be created. The plan shows the resource's attributes and the actions to be taken.

```
carlosdc@192 Lab1_Terraform % terraform init
carlosdc@192 Lab1_Terraform % terraform validate
Success! The configuration is valid.

carlosdc@192 Lab1_Terraform % terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

# random_string.suffix will be created
+ resource "random_string" "suffix" {
  + id          = (known after apply)
  + length      = 16
  + lower       = true
  + min_lower   = 0
  + min_numeric = 0
  + min_special = 0
  + min_upper   = 0
  + number      = true
  + numeric     = true
  + result      = (known after apply)
  + special     = true
  + upper       = true
}

Plan: 1 to add, 0 to change, 0 to destroy.
```

Note: You didn't use the `-out` option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform

Explorer

Lab1_Terraform

.terraform

.terraform.lock.hcl

main.tf

terraform.tfstate

main.tf

terraform.tfstate

```
1 {
2   "resources": [
3     {
4       "mode": "managed",
5       "type": "random_string",
6       "name": "suffix",
7       "provider": "provider[\"registry.terraform.io/hashicorp/random\"]",
8       "instances": [
9         {
10          "schema_version": 2,
11          "attributes": {
12            "id": "7zpz]Pe7m kn7GyF",
13            "keepers": null,
14            "length": 16,
15            "lower": true,
16            "min_lower": 0,
17            "min_numeric": 0,
18            "min_special": 0,
19            "min_upper": 0,
20            "number": true,
21            "numeric": true,
22            "override_special": null,
23            "result": "7zpz]Pe7m kn7GyF",
24            "special": true,
25            "upper": true
26          },
27          "sensitive_attributes": [],
28          "identity_schema_version": 0
29        }
30      ]
31    }
32  ]
33 }
```

Problems

Output

Debug Console

Terminal

Ports

```
carlosdc@192 Lab1_Terraform % terraform apply
+ result      = (known after apply)
+ special     = true
+ upper       = true
}

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

random_string.suffix: Creating...
random_string.suffix: Creation complete after 0s [id=7zpz]Pe7m(kn7GyF)

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
carlosdc@192 Lab1_Terraform %
```

Outline

Timeline